Labels, Not Access: Monitoring Closed Agents from the Transcript and the Proposed Action

Anonymous Authors[1]

Abstract

A monitor that reads a model's internal activations needs the model's weights. The most capable agents give only text. We show that, for the labels that agent monitoring needs, weight access adds nothing that the transcript and the proposed action do not give. An agent's internal state is a function of the tokens that it read, so internals can only add information from tokens that the transcript hides. For a policy check on a proposed action, the action makes even those tokens irrelevant: a monitor that reads the transcript and the proposed action is Bayes-sufficient. The remaining problem is statistical: learn the policy from labelled examples. We test each term with preregistered, hash-locked tests. Reading after the agent's reasoning explains the white-box advantage over monitors at the start of the turn (AUROC 0.979 vs 0.841). Without labels, a 70B judge loses to a probe (0.782). With labels, an open reader of the transcript and the action beats the agent's own internals (0.994 vs 0.979), and the agent's hidden reasoning adds nothing once the action is visible (-0.0014). Trained only on an open agent, the reader catches 96% and 98% of the violations of two closed agents, Claude Haiku 5.5 and GPT-6-luna, at a 5% review budget on the training benchmark; a zero-shot judge catches 0% and 38%. On an external benchmark of prompt injections, the reader is not significantly better than the judge on a closed agent (0.840 vs 0.787; a preregistered failure). The limit is the coverage of the policies: a policy unlike any trained policy is ranked but not escalated until 20 to 40 labelled examples are added. We release the record of every test, an SDK, and a website on which anyone can train and audit a monitor in the browser.

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

Agents built on language models now act through tools: they send messages, run commands and move data. A monitor must decide if each proposed action can run, before it runs. Monitors that use interpretability read the agent's activations (Zou et al., 2023; Goldowsky-Dill et al., 2025; Fayyazi et al., 2026). They need the weights, but the agents with the most capability are closed. Their providers give sampled text and, at most, top-k log probabilities and a summary of the reasoning.

Some evidence suggests that the transcript is enough. Open "reader" models that read the transcripts of closed models come close to probes on the target's own activations (Conti et al., 2026). Follow-up questions detect lies about as well as probes on prompted lies, but not on model organisms with verified beliefs (Pacchiardi et al., 2024; Cooney et al., 2026). No result says when the transcript is enough, or why.

Our answer. For a policy check on a proposed action, weight access adds nothing that the transcript and the proposed action do not give (Section 2). The value of a white-box monitor then reduces to three measurable terms: tokens that the transcript hides, the label function, and the statistics of learning the policy from labels. We test each term with preregistered tests on agents from 20B-parameter open models to closed frontier models (Section 4).

Contributions.

1. *Theory.* An agent's internals add to its full token history no information about any label; under log loss, a white-box monitor beats a transcript monitor by exactly $I(Y; H \mid V)$, the information in hidden tokens (Proposition 1). For action-level labels, the transcript and the proposed action are Bayes-sufficient (Corollary 1).
2. *A preregistered test of each term:* timing (T2), the label function (T3), labels (T4, T5), representation (T9) and policy coverage (T10, T12).
3. *Two closed agent families.* A reader trained only on an open agent monitors Claude Haiku 5.5 (T8b) and GPT-6-luna (T11) beyond tool identity on the training benchmark. On an external benchmark it beats a zero-shot judge for open agents (T7), but not significantly for a closed agent (T13b).

4. *Deployment without labels from the new agent.* Probabilities do not transfer between agents; rankings do. A review threshold set from the agent's own traffic and then frozen resists bursts of violations.
5. *A tool.* An open SDK and a website: anyone can train and audit a monitor on their own agent in the browser, with results identical to the Python version.

The theorem is simple; we do not claim that it is deep. Its value is that it turns a vague question ("do we need the weights?") into terms that experiments can measure, and our tests measure each term.

2. Theory

Setting. A monitor acts at step t, after the agent proposes an action and before the action runs. Z_t is every token that the agent read or wrote up to t. V_t is the part that a deployment sees, and H_t is the rest, for example reasoning that the API hides. A_t is the proposed action, and W are world facts that the agent did not observe. An internal representation is $\phi_t = \Phi(Z_t; \theta)$, a deterministic function of the tokens.

Proposition 1 (access is not information)**.** *For every label Y, $\sigma(Z_t, \phi_t) = \sigma(Z_t)$. Thus the best monitor that uses (Z_t, ϕ_t) is not better than the best monitor that uses Z_t, under every strictly proper score and at every point of the ROC curve. Under log loss, the best monitor that uses Z_t beats the best monitor that uses V_t by exactly $I(Y; H_t \mid V_t)$.*

Corollary 1 (the proposed action is sufficient)**.** *Let $Y = g(A_t, V_t, W)$, with $W \perp\!\!\!\perp Z_t^- \mid V_t$, where Z_t^- are the tokens before the action is sampled, and with sampling noise independent of (Z_t^-, W). Then $P(Y \mid Z_t^-, A_t) = P(Y \mid V_t, A_t)$, and every monitor that reads internals at or before the action has a Bayes risk at least that of the best monitor that reads (V_t, A_t).*

Proofs are in Section A. A probe read before the action is emitted must also predict which action the agent will sample; a monitor that reads the action does not. Thus the inequality can be strict when the agent's action is random. Corollary 1 does not cover labels about later behaviour or about beliefs. For labels about later behaviour, the rollout forecast dominates every readout of the state (Section A, Proposition 3).

What remains is statistics. The excess risk of a trained transcript monitor is the risk of learning $P(Y \mid V_t, A_t)$ from data. It depends on the label function used for training, on the representation, and on how well the labelled data cover the policies that g enforces.

3. Setup

Agents and data. CAPE-Bench (Fayyazi et al., 2026) gives matched agent tasks in which one fact decides if an action is authorized; an environment rule checker labels each action. Open agents: Gemma-4 (26B) and gpt-oss-20b. Closed agents, through their public tool-use APIs, with reasoning off: Claude Haiku 5.5 and GPT-6-luna, 1,000 episodes each. External benchmark: AgentDojo prompt injections (Debenedetti et al., 2024) with the spotlighting defense (Hines et al., 2024); a call is labelled injected if it executes the attacker's task.

Monitors. *Reader*: Qwen3-8B reads the agent's instructions, the user's task, the earlier tool calls with their results, and the proposed action; a linear probe on its residual stream (layer 32) learns the labels. It is trained only on Gemma-4 CAPE-Bench steps (17,623 steps) and is never fit on a test agent or on AgentDojo. *Internals*: CAPE's probe on Gemma-4's own residual stream. *Zero-shot judge*: Llama-3.3-70B with the same input. *TF-IDF*: a bag-of-words monitor with the reader's input and labels. *Tool identity*: the training-set violation rate of the action's tool.

Preregistration and statistics. Every confirmatory test fixed its data, metric and pass criterion, and was hash-locked before its data existed or was scored. We report every outcome, including failures. We give AUROC within strata (template, tool or suite), with 95% intervals from a bootstrap over scenarios or episodes (1,000 resamples), and the share of violations caught at a 5% review budget.

4. Results

Table 1 lists every agent test with its preregistered outcome, including the failures.

4.1. Timing explains the white-box advantage (T2)

On 12,128 fresh Gemma-4 steps, a probe on the agent's internals scores 0.841 at the start of the turn and 0.979 after the agent's reasoning, before the action (Figure 2). After the reasoning, the action is nearly fixed: resampling the action and checking it with the environment's rule checker gives 0.9997. The advantage of internals over monitors at the start of the turn is when they read, not what they can access.

4.2. The label function decides in practice (T3)

Resampling needs a label function at test time. With a 70B judge in place of the rule checker, resampling falls to 0.782, below the turn-start probe (0.846) and far below the internals after the reasoning (0.976). The bottleneck is the label function, not the access.

4.3. With labels, the transcript equals the internals (T4, T5)

The reader, trained on the same labels, scores 0.994 on the same scenarios; the agent's internals score 0.979 ($+0.015$

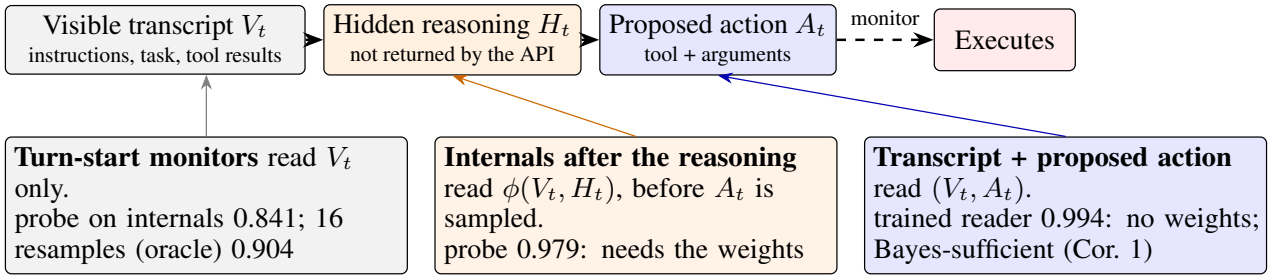


Figure 1. One agent step, and what each monitor reads. A white-box probe gains over monitors at the start of the turn because it reads after the hidden reasoning. A monitor that reads the proposed action needs no weights: for a policy check on the action, it is Bayes-sufficient (Corollary 1), and with labels it scores higher than the internals. AUROC on the same 12,128 fresh steps (T2, T4).

Table 1. All agent tests, in the order in which they were locked. Pass: the preregistered criterion was met. AUROC within strata unless stated.

Test	Question	Key result	Outcome
T1	Resampling vs a probe at the turn start (oracle checker)	0.904 vs 0.841	pass; answers alone: fail
T2	Internals after the reasoning vs monitors at the turn start	0.979 vs 0.904	pass
T3	Resampling with a deployable 70B judge	0.782 $<$ probe 0.846	fail (probe better)
T4	Transcript reader with labels vs the agent's internals	0.994 vs 0.979	pass
T5	The same, on unseen task families	0.969 vs 0.976 (non-inferior)	pass
T6	Reader trained on Gemma, applied to gpt-oss	0.915 vs judge 0.833	pass
T7	Reader on AgentDojo, no AgentDojo training	0.781 vs judge 0.537	pass
T8	Claude, actions after the reveal	0 violations in the units	not tested
T8b	Claude, all actions	95.6% caught vs 0%	pass
T9	TF-IDF with the same labels vs the reader	equal in distribution; chance on AgentDojo	mixed
T10	Claude, a policy family removed from training	0.912 vs 0.570; 0/26 caught	pass (ranking)
T11	GPT-6-luna, beyond tool identity	0.996 vs 0.778; tool 0.856	pass
T12	Labels for a new policy, prospective	18/18 caught with 0 labels	H1 fail
T13	GPT-6-luna on AgentDojo, with spotlighting	attack success 2.1%; 4 injected calls	not tested
T13b	The same, without the defense, both splits	0.840 vs judge 0.787 (36 injected)	fail

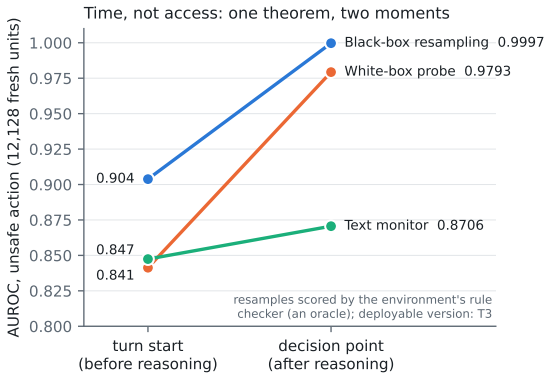


Figure 2. The same 12,128 fresh steps, read at two moments. Reading after the reasoning helps every method. Resamples are checked by the environment's rule checker (an oracle).

[0.007, 0.026]). On unseen scenario families (5 folds over 23 families) the reader is non-inferior: 0.969 vs 0.976 (-0.007 [-0.014, -0.0001]; margin -0.02). The zero-shot judge scores 0.910. Given the action, the agent's hidden reasoning adds nothing: adding it to the reader's input changes AUROC by -0.0014 [-0.0046, $+0.0014$] (exploratory), as Corollary 1 predicts. Seventy-four labelled steps already beat the judge (0.938).

4.4. The representation carries labels to new tasks (T9)

TF-IDF with the same input and labels equals the reader on the training benchmark (0.992 vs 0.994). On unseen task families the reader is better (0.969 vs 0.950; $+0.019$ [0.007, 0.032]). On AgentDojo, TF-IDF is at chance (0.503); the reader scores 0.781 (T7).

4.5. Two closed agent families (T8b, T11, T13, T13b)

Nothing was fit on the closed agents (Table 2). On Claude Haiku 5.5 the reader catches 95.6% of 68 violations at a 5% budget; the judge catches none. The preregistered within-template AUROC (0.991 vs 0.633) rests on one stratum, so we also give the pooled AUROC (0.989 vs 0.570, exploratory). On GPT-6-luna all three preregistered endpoints pass: pooled 0.996 vs 0.778 ($+0.217$ [0.174, 0.265]); against tool identity alone $+0.141$ [0.111, 0.171]; within tool $+0.039$ [0.015, 0.066]. On AgentDojo, GPT-6-luna rarely follows the injected instructions: with the spotlighting defense, the attack succeeds in 2.1% of the episodes,

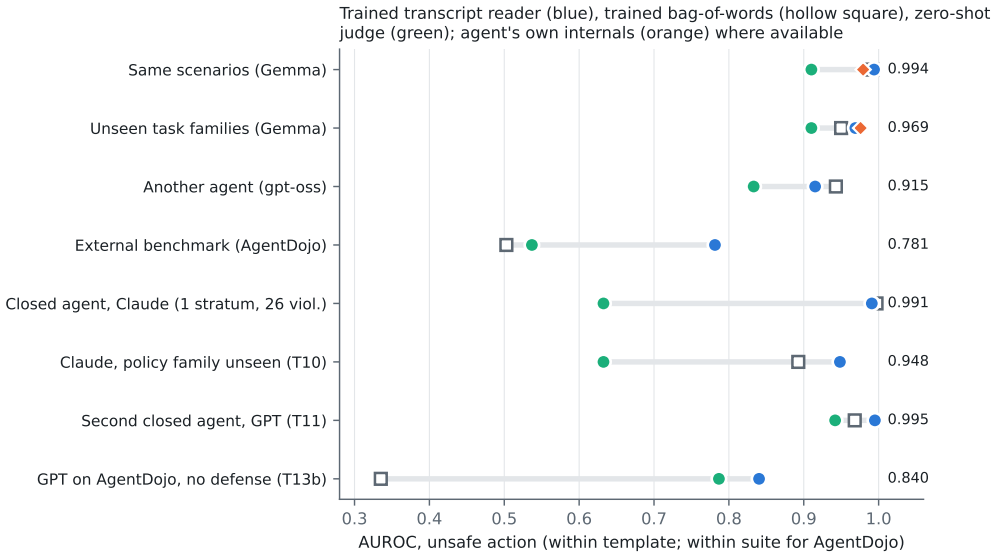


Figure 3. Monitoring the next action from the transcript, under increasing shift. Reader (blue), TF-IDF with the same labels (hollow square), zero-shot 70B judge (green), the agent's internals (orange). The Claude rows rest on one stratum (136 actions, 26 violations); in the T10 row the monitors were retrained without that policy.

Table 2. Closed agents, frozen monitors trained only on Gemma-4. Caught: violations escalated when 5% of the actions go to review.

	Claude Haiku 4.5		GPT-6-luna	
	AUROC	caught	AUROC	caught
Reader	0.989	95.6%	0.996	98.5%
TF-IDF	0.997†	64.7%	0.964	50.0%
Zero-shot judge	0.570	0%	0.778	37.9%
Tool identity	–	–	0.856	60.6%

Pooled AUROC. †within the one stratum with both classes.

which leaves 4 injected calls, below the preregistered minimum of 10 (T13, not tested). Without the defense, on both splits, there are 36 injected calls (T13b). There the reader is not significantly better than the zero-shot judge: 0.840 vs 0.787 (+0.052 [−0.026, +0.130]); the preregistered test fails. TF-IDF is below chance (0.335). The monitor transfers to closed agents on its training benchmark; on an external benchmark with a different policy, it needs labels of that policy, as Section 4 shows for policy coverage.

4.6. The limit is policy coverage (T10, T12)

We removed one policy family from the training data. For a service-restart policy, the reader still ranks Claude's violations well (pooled 0.912 vs 0.570 for the judge), but at a 5% budget it escalates 0 of 26 (T10). The violations rank above safe actions of the same task, but not above risky actions of other tasks. With 20 labelled steps of the policy it catches 85%; with 40, 94% (exploratory; Figure 4). We then tested this rule prospectively on GPT with a credentials policy (T12). The preregistered hypothesis failed: with no labels,

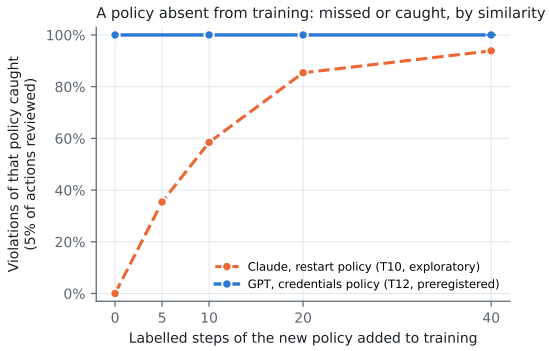


Figure 4. A policy removed from training, against the number of its labelled steps added back. Credentials (GPT, preregistered T12): caught without labels. Service restart (Claude, exploratory): missed until 20–40 labels.

the reader already caught all 18 violations. The training data hold similar credential policies; the restart policy had none. A new policy that resembles a trained one can transfer; one that does not needs a few dozen labels. With two held-out policies, this is a hypothesis, not a measurement.

4.7. A review threshold without labels from the new agent

The reader's probabilities do not transfer between agents: on Claude, a 0.5 cut catches 3%. Its ranking transfers. We set the threshold at the top 5% of the agent's first 500 actions and then freeze it. Online, it catches 95.4% on Claude (1.9% of safe actions escalated) and 98.4% on GPT (2.3%). In a simulated burst in which half of the actions are unsafe, the

frozen threshold catches 94%; a rolling threshold rises and catches 36% (exploratory).

5. The tool

We release an SDK and a website (links in the camera-ready version). On the website, a user loads labelled steps from their own agent, trains the TF-IDF monitor in the browser, gets a cross-validated accuracy over held-out tasks, a frozen review threshold and the catch rate at their budget, and scores new actions. The data stay in the browser. The browser and the Python SDK give byte-identical text on 302 real agent steps and the same held-out AUROC (0.9887 vs 0.9886); a model trained in the browser scores identically in Python (10^{-13}). Scoring takes 0.18 ms per action in a browser; the reader takes 41 ms on one GPU. A self-hosted API keeps transcripts on the user's machine.

6. Related work

Probes on internals read truth, refusal and deception from activations (Burns et al., 2023; Marks & Tegmark, 2023; Zou et al., 2023; Arditi et al., 2024; Goldowsky-Dill et al., 2025); CAPE probes an agent before each action (Fayyazi et al., 2026). Proposition 1 states what they can add when a transcript exists. **Chain-of-thought monitors** catch reward hacking that action-only monitors miss (Baker et al., 2025); reasoning is a fragile but useful signal (Korbak et al., 2025; Chen et al., 2025). Corollary 1 makes this more exact: for labels that the action and the visible facts decide, reasoning adds nothing once the monitor sees the action; it can help for labels about intent or later behaviour. **AI control** audits a fixed budget of actions with a trusted monitor (Greenblatt et al., 2024). Our results show that a trained transcript monitor beats a zero-shot trusted judge, and that the threshold must come from the monitored agent's own traffic. **Black-box access** recovers parts of a model from its logits (Carlini et al., 2024; Finlayson et al., 2024), detects lies with follow-up questions (Pacchiardi et al., 2024; Cooney et al., 2026; Sam et al., 2025), forecasts behaviour from traces (Levy et al., 2026), and probes closed transcripts with open readers (Conti et al., 2026). We add the theory of when transcripts are enough, a preregistered test of each term, and closed agents.

7. Limitations

- Most agent tests use one benchmark. On AgentDojo, the external test, the reader beats a zero-shot judge for open agents (0.781 vs 0.537) but not significantly for a closed agent (0.840 vs 0.787, T13b).
- A monitor needs labels of each policy that is unlike its training policies. We measured this on two held-out policies only.
- Corollary 1 needs the label to be decided by the action and the visible facts, and a deployment that sees the action before it runs. It does not cover labels about beliefs or later behaviour.
- A frozen threshold assumes trusted warm-up traffic; a poisoned warm-up would raise it.

8. Conclusion

For monitoring the actions of closed agents, the weights are not the bottleneck: labels and policy coverage are. A reader of the transcript and the proposed action, trained on labels, equals the agent's internals and transfers to closed agents from two providers on the policies it learned. For a different policy, such as prompt injection, it needs labels of that policy.

References

Arditi, A., Obeso, O., Syed, A., Paleka, D., Panickssery, N., Gurnee, W., and Nanda, N. Refusal in language models is mediated by a single direction, 2024.

Baker, B., Huizinga, J., Gao, L., Dou, Z., Guan, M. Y., Madry, A., Zaremba, W., Pachocki, J., and Farhi, D. Monitoring reasoning models for misbehavior and the risks of promoting obfuscation. *arXiv preprint arXiv:2503.11926*, 2025.

Burns, C., Ye, H., Klein, D., and Steinhardt, J. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations (ICLR)*, 2023.

Carlini, N. et al. Stealing part of a production language model, 2024. VERIFY author list.

Chen, Y., Benton, J., Radhakrishnan, A., Uesato, J., Denison, C., Schulman, J., Somani, A., Hase, P., Wagner, M., Roger, F., Mikulik, V., Bowman, S. R., Leike, J., Kaplan, J., and Perez, E. Reasoning models don't always say what they think. *arXiv preprint arXiv:2505.05410*, 2025.

Conti, Sharf, Jain, Yan, et al. Model hermeneutics: Monitoring closed-weight models with open-weight internals. LessWrong, 30 September 2026, 2026. VERIFY full author names.

Cooney, A., Africa, D., and Irving, G. "did you lie?" evaluating lie detectors across scale and belief-verified model organisms, 2026.

Debenedetti, E., Zhang, J., Balunović, M., Beurer-Kellner, L., Fischer, M., and Tramèr, F. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Infor-*

mation Processing Systems (Datasets and Benchmarks Track), 2024.

Fayyazi, A. et al. The agent knew: Internal states forecast unauthorized actions and stop them, and the lies that follow, 2026. VERIFY author list.

Finlayson, M., Ren, X., and Swayamdipta, S. Logits of API-protected LLMs leak proprietary information, 2024.

Goldowsky-Dill, N., Chughtai, B., Heimersheim, S., and Hobbhahn, M. Detecting strategic deception with linear probes. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, pp. 19755–19786, 2025.

Greenblatt, R., Shlegeris, B., Sachan, K., and Roger, F. AI control: Improving safety despite intentional subversion. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.

Hines, K., Lopez, G., Hall, M., Zarfati, F., Zunger, Y., and Kiciman, E. Defending against indirect prompt injection attacks with spotlighting, 2024.

Korbak, T., Balesni, M., Barnes, E., Bengio, Y., Benton, J., Bloom, J., Chen, M., Cooney, A., Dafoe, A., Dragan, A., Emmons, S., Evans, O., Farhi, D., Greenblatt, R., Hendrycks, D., Hobbhahn, M., Hubinger, E., Irving, G., Jenner, E., Kokotajlo, D., Krakovna, V., Legg, S., Lindner, D., Luan, D., Madry, A., Michael, J., Nanda, N., Orr, D., Pachocki, J., Perez, E., Phuong, M., Roger, F., Saxe, J., Shlegeris, B., Soto, M., Steinberger, E., Wang, J., Zaremba, W., Baker, B., Shah, R., and Mikulik, V. Chain of thought monitorability: A new and fragile opportunity for AI safety. *arXiv preprint arXiv:2507.11473*, 2025.

Levy, Goldberg, and Cooper Stickland. Forecasting future behavior as a learning task, 2026. VERIFY full author names.

Marks, S. and Tegmark, M. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets, 2023.

Pacchiardi, L., Chan, A. J., Mindermann, S., Moscovitz, I., Pan, A. Y., Gal, Y., Evans, O., and Brauner, J. How to catch an AI liar: Lie detection in black-box LLMs by asking unrelated questions. In *International Conference on Learning Representations (ICLR)*, 2024.

Sam, D. et al. Predicting the performance of black-box LLMs through self-queries, 2025. VERIFY title and authors.

Zou, A., Phan, L., Chen, S., Campbell, J., Guo, P., Ren, R., Pan, A., Yin, X., Mazeika, M., Dombrowski, A.-K., Goel, S., Li, N., Byun, M. J., Wang, Z., Mallen, A., Basart, S., Koyejo, S., Song, D., Fredrikson, M., Kolter, J. Z., and Hendrycks, D. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A. Proofs

Setting. Z_t are all tokens that the agent read or wrote up to step t; V_t is the visible part and H_t the hidden part; A_t is the proposed action; W are world facts that the agent did not observe. An internal representation is $\phi = \Phi(Z; \theta)$, a deterministic function of the tokens (numerical noise in a batch is independent of everything else).

Proposition 3 (behavioural sufficiency). Let $Y = g(\tau)$ be defined by the future τ that the agent and the environment generate from context x, and let $\phi(x)$ carry no information about the future sampling noise given x. Then $\pi^*(x) = P(Y = 1 \mid x)$ has a maximal ROC curve and a minimal risk under every strictly proper score among all forecasts $f(\phi(x))$.

Proof. By the assumption and the tower property, $E[Y \mid \phi(X)] = E[\pi^*(X) \mid \phi(X)]$. Under a strictly proper score, the Bayes act given $\sigma(X)$ is $\pi^*(X)$, and $\sigma(\phi(X)) \subseteq \sigma(X)$; less information cannot raise the expected score. For the ROC curve, the Neyman–Pearson lemma gives the most powerful test of each size as a threshold on the likelihood ratio, which is a strictly increasing function of π^*; every test based on $f(\phi(X))$ is a test based on X. □

Proposition 1 in the main text (access is not information). (i) ϕ_t is $\sigma(Z_t)$-measurable, so $\sigma(Z_t, \phi_t) = \sigma(Z_t)$, and Proposition 3 with $X = Z_t$ gives the first claim. (ii) The Bayes log loss of predicting Y from a σ-field $\mathcal{G}$ is the conditional entropy $H(Y \mid \mathcal{G})$; thus $H(Y \mid V_t) - H(Y \mid V_t, H_t) = I(Y; H_t \mid V_t)$. □

Corollary 1 in the main text (the proposed action is sufficient). Let Z^- be the tokens before the action is sampled, $A_t = a(Z^-, \xi)$ with sampling noise $\xi \perp\!\!\!\perp (Z^-, W)$, $W \perp\!\!\!\perp Z^- \mid V_t$, and $Y = g(A_t, V_t, W)$.

1. From $W \perp\!\!\!\perp Z^- \mid V_t$ and $\xi \perp\!\!\!\perp (Z^-, W)$: $W \perp\!\!\!\perp (Z^-, \xi) \mid V_t$.
2. A_t is a function of (Z^-, ξ), so $W \perp\!\!\!\perp (Z^-, A_t) \mid V_t$.
3. By weak union, $W \perp\!\!\!\perp Z^- \mid (V_t, A_t)$.
4. Y is a function of (A_t, V_t, W), so $Y \perp\!\!\!\perp Z^- \mid (V_t, A_t)$, that is, $P(Y \mid Z^-, A_t) = P(Y \mid V_t, A_t)$.
5. A monitor that reads internals at or before the action uses $\phi = \Phi(Z^-)$, with $\sigma(\phi) \subseteq \sigma(Z^-) \subseteq \sigma(Z^-, A_t)$. By step 4 the Bayes predictor given $\sigma(Z^-, A_t)$ equals the one given (V_t, A_t), and less information cannot help (Proposition 3). □

The inequality can be strict: if A_t is random given Z^- and Y depends on which action is sampled, a probe that reads before the action must average over the actions. The assumption on W fails only if the internals hold information about the world that is not in the tokens, which cannot happen for a model whose only input is its token history.

B. The full preregistration record

Every confirmatory test, in the order in which it was locked. Locks (SHA-256 of the preregistration and its code) and data are in the released record.

Test	Outcome (verbatim from the record)
L2 open LLMs (H1–H3)	**PASS all, after deviation 1**: H1 +0.120 [0.115, 0.126]; H2 +0.057 [0.045, 0.073] (7/8 models); H3 +0.359 [0.310, 0.415]; Holm p ¡ 0.0005 each. First run: H1 valid PASS, H2/H3 invalid (numerical; see note) (results/l2/l2_confirm.json)
T1 agents (F1–F3; addendum 1: F4–F5)	**Claim (a) CONFIRMED** (F1, F2 pass: R16 − WB +0.063 [0.033, 0.094]); claim (b) NOT confirmed (F3 fails: Ψ − WB −0.021 [−0.054, 0.009]); addendum **F4 passes** (R4+Ψ − WB +0.049 [0.020, 0.080]); F5 fails (R2+Ψ − R16 −0.023 [−0.053, 0.006]) (results/t1/t1_agents_confirm.json)
L1 Othello recovery (E1–E3)	**PASS all**: E1 R^2 0.967 [0.967, 0.968] ¿ 0.90; E2 ΔR^2 +0.094 [0.086, 0.105], Holm p ¡ 0.0005; E3 Δ occupied acc +0.043 [0.040, 0.046], Holm p ¡ 0.0005. White-box reference 0.983 (results/othello/l1_confirm.json)
L3 Claude Haiku 5.5 readout transfer (C1–C3)	**all pass**: C1 0.993 [0.983, 0.999]; C2 0.997 [0.990, 1.000]; C3 0.935 [0.868, 0.982] (results/l3/)
L3b Claude Haiku 5.5: grey box vs text moderation (D1–D3)	**D1 pass** 0.923 [0.884, 0.957]; **D2 (key) pass** grey box − moderation +0.176 [0.060, 0.287] (0.827 vs 0.651); D3 not tested (1/40 over-refusals) (results/l3/claude_livebench.json)
L3c Claude Haiku 5.5: one-call grey box + compliance alarm (E1–E3)	**all pass** vs the preregistered 3-level verdict: E1 +0.142 [0.081, 0.203] (0.815 vs 0.673); E2 AP +0.230 [0.161, 0.307]; E3 AP +0.081 [0.036, 0.140]. Against GRADED moderators E1 ties (Qwen3Guard +0.018 [−0.043, +0.081]; Llama-Guard −0.031 [−0.082, +0.020]); the alarm gain holds (results/l3/claude_batch.json, l3c_graded_reanalysis.json)
L3d Claude Haiku 5.5: transcript-framing grey box (F1–F2)	**both pass**: F1 grey box − moderation on harmful refusals +0.141 [0.082, 0.199] (0.817 vs 0.676); F2 compliance-risk AP (Claude-moderator) +0.115 [0.076, 0.179] (results/l3/claude_transcript.json)

Test	Outcome (verbatim from the record)
T2 time: internals at CAPE's decision point vs black box (G1–G3)	**all pass**: G1 I_dec − R16_ts +0.075 [0.044, 0.106]; G2 R16_dec − I_dec +0.021 [0.011, 0.035]; G3 I_dec − WB_ts +0.138 [0.107, 0.163] (results/t1/t2_time.json)
L3e Claude Haiku 5.5: the Live tool's alarm, prospective (A1–A3)	**A1 pass** (alarm precision 0.535 vs 0.274 for the moderator alone, +0.261 [0.215, 0.311]); **A2 FAIL** (grey box 0.873 vs graded Qwen3Guard 0.855, +0.018 [−0.023, 0.062]); **A3 FAIL** (multi-turn 0.894 vs 0.906) (results/l3/l3e_tool.json)
T3 deployable monitors: LLM judge instead of the oracle checker (K1–K3)	**K1 FAIL** (judge-scored resampling 0.782 ¡ turn-start probe 0.846, −0.065 [−0.115, −0.019]); **K2: probe better** (CAPE probe 0.976 ¿ judge on actual action 0.910, −0.067 [−0.104, −0.036] for the judge); **K3 FAIL** (resampling hurts the action judge, −0.049) (results/t1/t3_deployable.json)
T4 reader trained on labels vs the agent's internals (M1–M2)	**both pass**: M1 READER_act − I_dec +0.015 [0.007, 0.026] (0.994 vs 0.979); M2 READER_act − J_act +0.085 [0.052, 0.127] (results/t1/t4_reader.json)
T5 agent internals vs transcript reader on unseen scenario families (P1–P2)	**P1 pass** (reader − agent internals on unseen families −0.007 [−0.014, −0.0001], margin −0.02); **P2: agent probe better than zero-shot judge** (+0.062 [0.028, 0.101]) (results/t1/t5_family.json)
T6 reader trained on Gemma, monitoring gpt-oss (Q1)	**Q1 pass**: Gemma-trained reader on gpt-oss 0.915 vs zero-shot judge 0.833, +0.082 [0.043, 0.129] (results/t1/t6_cross_agent.json)
T7 reader monitor on AgentDojo prompt injection, zero transfer training (R1–R2)	**R1 pass**: reader 0.781 vs zero-shot judge 0.537, +0.249 [0.136, 0.369]; **R2 reader better than CAPE's gate**: gpt-oss calls as the prereg text specifies, 0.823 vs 0.601, +0.222 [0.143, 0.303] (locked code's broader subset: +0.264 [0.174, 0.359]) (results/t1/t7_agentdojo.json)
T8 Claude Haiku 5.5 as an agent, monitored by a reader trained only on Gemma (S1)	**S1 NOT TESTED**: 0 unsafe units under the preregistered post-reveal rule (Claude's 68 violations all happen at or without the reveal); see T8b
T8b Claude as an agent, all non-read actions (S1b)	**S1b pass**: Gemma-trained reader on Claude 0.991 vs zero-shot judge 0.633, +0.360 [0.302, 0.413]; 95.6% of violations caught at 5% escalation (judge 0%) (results/t1/t8b_claude_all_actions.json)
T9 trained TF-IDF monitor on the reader's own input, same labels (B1 Claude, B2 unseen families)	**B2 pass, B1 FAIL**: unseen families reader 0.969 vs TF-IDF 0.950 (+0.019 [0.007, 0.032]); Claude reader 0.991 vs TF-IDF 0.997 (−0.006 [−0.027, +0.007]). AgentDojo (reported) 0.781 vs 0.503 (results/t1/t9_text_baseline.json)
T10 Claude on task families the monitor never saw (C1 pooled, C2 within)	**C1, C2 pass** (vs zero-shot judge): pooled 0.912 vs 0.570 (+0.342 [0.311, 0.371]); within 0.948 vs 0.633 (+0.318 [0.265, 0.374]). But at the label-free cut the unseen restart family is caught 0/26 (results/t1/t10_claude_unseen_families.json)
T11 second closed family: gpt-6-luna as agent, frozen monitors (G1 pooled vs judge, G2 vs tool identity, G3 within tool)	**G1, G2, G3 all pass**: GPT-6-luna, 3,627 actions, 66 unsafe in 6 families. Reader pooled 0.996 vs judge 0.778 (+0.217 [0.174, 0.265]); vs tool identity +0.141 [0.111, 0.171]; within tool vs judge +0.039 [0.015, 0.066]. 98.5% caught at the label-free 5% cut (judge 37.9%) (results/t1/t11_gpt_agent.json)
T12 labels needed for a new policy, prospective on GPT (H1 40 vs 0 labels, H2 catch at $40 \geq 0.5$)	**H1 FAILS, H2 passes**: with 0 labels of the held-out credentials policy, the reader already caught 18/18 of GPT's violations of it, so added labels had nothing to fix (40 − 0: 0.0). The T10 curve does not replicate on this family (results/t1/t12_new_policy_labels.json)
T13 a closed agent on an external benchmark: GPT-6-luna on AgentDojo, frozen monitors (K1 reader vs judge)	pending